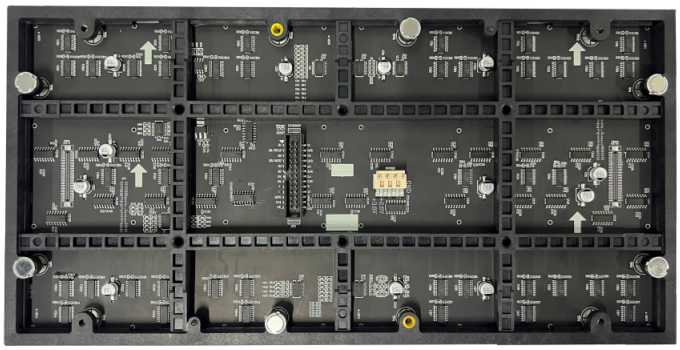
BK-COB P1.53
Specifications

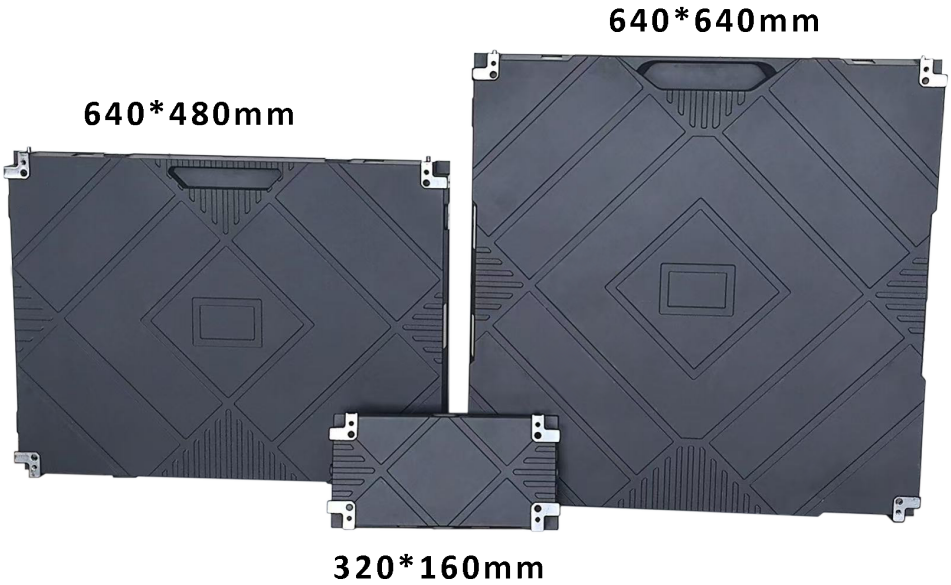
Approve	Check	Formulate
Version: V1.0		
Date: 2025-06-03		

Shenzhen BAKO Optoelectronics Co.,Ltd

I. Module Picture



II. Panel Picture



III. Parameters

BK-COB P1.53 Specification		
LED Parameters	Pixel Configuration	RGB 3-in-1
	Packaging Technology	Flip Chip COB
COB Module Parameters	Pixel Pitch（mm）	1.53
	Module Size（W×H mm）	320*160
	Module Resolution（W×H）	208×104
	Scanning Method	1/104
	Receiving Card Interface	HUB320F
	Circuit Board	Anode Circuit
	Surface Processing	Matte
	Surface Hardness	4H
Panel Parameters	Panel Resolution (W*H)	416*208
	Panel Size (W*H*D mm)	640*480
	Pixel Density (dots/㎡)	422500
	Panel Flatness(mm)	<0.1
	Panel Weight(With Module) (Kg)	7
	Lamp Board Connection Mode	Cable connection
	Installation/Maintena-nce Method	Front service
	Protection Grade	Front IP54/Rear IP43
	Brightness after calibration(cd/㎡)	600
	LED Chip Life Span（h）	100,000
Image Parameters	Brightness Uniformity (after calibration)	≥97%
	Color Uniformity (after calibration)	Within ±0.001Cx,Cy
	Refresh Rate (Hz)	3840
	Frame Frequency	50/60Hz
	Color Temperature	3000K-15000K adjustable
	Grayscale (bit)	14
	Horizontal/Vertical Viewing Angle(H/V)	≥175°/175°
	Contrast Ratio	15000:1

Electrical Parameters	Input Voltage(AC V)	100-240V
	Max. Power Consumption (W/㎡)	380
	Average Power Consumption (W/㎡)	110
Control System	Receiving Card	MRV532
Environment Parameters	Working Temperature /Humidity（℃/RH）	-10℃~+40℃/10%RH~90%RH, No Frosting
	Storage Temperature /Humidity（℃/RH）	-20℃~+60℃/10%RH~90%RH, No Frosting

IV. Product Usage Instruction

Item	Description
Temperature Requirement	Storage Temperature:－20℃～＋60℃, need to cool down when over 60℃. Working Temperature:－10℃～＋40℃, exceed this temperature range, need to add temperature control equipment. The surface temperature of LED screen during operation should be: ≤60℃, exceed this range, should add temperature control equipment.
Humidity Prevention Requirement	Storage Humidity: 10%RH～90%RH, exceed 90%RH, need dehumidification treatment. Working Humidity: 10%RH～90%RH, exceed this range, need dehumidification treatment.
Handling of Overdue Storage	If store more than one month, should do 6hrs aging process before usage. The way of aging process: 10% brightness setting for 1hrs; 30% brightness setting for 1hrs; 60% brightness setting for 2hrs; 80% brightness setting for 1hrs;100% brightness setting for 1hrs. (Brightness gradually increases for aging test)
Dust-proof Requirement	Products should not be exposed to environments with a lot of dust, such as studio decoration and renovation. Special protection measures need to be taken for the display screen.
Protection Requirement from Corrosion gas	Corrosive gases, when present in the air or in environments containing salt or acidic gases, can cause phenomena such as corrosion of electronic components and crystalline leakage current.
Protection Requirement from electromagnetic radiation	The display should not be placed in an environment where electromagnetic radiation or radio frequency radiation exceed the interference level of 5 V/m compared to the electric field strength.
Shockproof Requirement	The display should be installed on a sturdy and reliable structure, which is free from strong vibrations.
Away from Water	The display is for indoor usage, not waterproof. Water can cause short-circuit or damage of the electric components.
Prevention from	Prevent the damage to electronic devices caused by static electricity in humid

static & lightning	environments, and also avoid the harm to the human body caused by leakage current or lightning.
Aware of personal Injury	The angle and height of the display installation should be appropriate. Sharp corners need to be protected to prevent harm to the human body from the hard shell.
Safety Requirements	It is necessary to ensure the stability of the power supply system, maintain normal levels of frequency, voltage and current harmonics. When install the display, the metal casing should be grounded together with the metal structure, and the overall grounding resistance of the display system should not exceed 1Ω, with the leakage current to ground not exceeding 5% of the input current. The electrical withstand strength should comply with the provisions of SJ/T 11141—2017 General Specification for Light Emitting Diode (LED) Display Screens 5.7.5.
Other points	The surface of the display should not be wiped with alcohol or other organic solvents. Instead, it should be cleaned with pure water and a fine, dust-free cloth. The surface of the display panel should not be touched directly with hands. When disassembling or assembling, anti-static gloves must be worn.

V.Product Installation Precautions

1. Guiding Principles

1) Make sure to install the display based on the reading and understanding of the user manual.

2) Under any circumstances, products must not be repaired by personnel who are not authorized by the factory;

3) The installation site must be stable and must not experience any sinking, tilting or falling. Moreover, the environment should not have any conditions such as excessive weight, radioactivity, pollution, corrosion or toxic gases.

4) All components must be provided by the original manufacturer or by suppliers certified by the original manufacturer;

5) Without the prior consent of the original manufacturer, it is strictly prohibited to

repair or replace the components at will.

6) Please operate according to the product instructions. If you encounter any problems, please contact our company.

2. Safety Instructions

2.1 Safety of Personnel

1) When performing work at heights, appropriate protective measures must be taken. Workers should wear qualified safety helmets and safety belts and other necessary safety protection equipment.

Anti-static Gloves	Safety Helmet	Safety Rope

2) The single panel weighs approximately 4kgs. During the operation, great attention must be paid to personal safety. Tools and materials must not be thrown. In case of cross-operation between the upper and lower parts, isolation facilities must be set up in the middle.

2.2 Product

1) During the transportation or storage process of all products, they must be properly protected and well-packaged. There should be no excessive external pressure on the products.

2) The product must not be exposed to rainwater before or during installation, and it should be operated in a dry and clean environment.

3) All components must not be stepped on, knocked or dropped. Please follow the instructions when moving or transporting the product.

4) Flammable substances: Keep flammable substances away from the equipment. During equipment operation, a large amount of energy is converted into heat. The operation of the equipment requires a certain amount of air flow to prevent the heat from endangering safe operation. Therefore, appropriate ventilation must be provided, and in necessary cases, air conditioners for cooling should be installed at the rear of the equipment.

2.3 Installation

1) When installing or removing the LED module, an suction cup tool is required, and the module should be disassembled and assembled vertically with the screen. At the same time, attention should be paid to the four edges of the lamp board, and they must not collide with the edges of other lamp boards. Also, be sure to take precautions against the lamp board from falling.

A. When fixing the cabinet, make sure the support frame is perpendicular to the installation plane; otherwise, the cabinet may not be installed properly.

B. When locking the axial screws, do not tighten them too much, otherwise it will cause the cabinet to bend, damaging the structure and flatness of the cabinet.

C. When the flatness between the cabinets is insufficient, the locking screws between the cabinets can be adjusted to regulate the flatness of the cabinets.

D. When installing the LED module, do not force it into place. This may cause dead lamps at the edges of the LED Module.

E. When installing the LED Module, it is necessary to ensure that the internal magnetic screws are tightened properly to guarantee that the height of the fixing points of the module is consistent.

2) When installing or removing the LED module, it is necessary to wear anti-static gloves.

3) After installation, the screen surface can be wiped clean with a lint-free cloth and pure water. Do not use any other solvents for cleaning.

2.4 Power Supply

1）Power supply and distribution cabinets are provided based on power consumption. All distribution systems must have an enclosure protection and comply with local circuit safety standards.

2）The power distribution system is installed close to the display, the power cord and data cable should not be pulled or damaged in any way.

3）Confirm whether the voltage input to the screen body is compatible with the screen's power supply. Please ensure the settings are correct before connecting the power cord.

4）Do not attempt to install damaged wires. Please replace them with new ones.

5）When the display is powered on, the current is relatively high for a moment. Choose an appropriate air switch.

2.5 Ground Connection

The display must be grounded using a separate grounding wire.

2.6 Attention during the Usage

1) The display must not be subjected to excessive external pressure or impact; otherwise, it will be damaged.

2) When cleaning the surface of the display, please follow the guidelines. During the cleaning process, only soft, dust-free cloths should be used.

3) The power must be cut off before removing the display.

4) All the accessory designs are only applicable for the installation and connection of this type of LED display.

5) No modification or duplication of any component is allowed. This display is made with special materials and through specific production procedures to ensure the strength of the components.

6) Make sure to strictly follow the installation instructions. If you have any questions regarding the safe application, you can consult the factory. The manufacturer shall not be held responsible for any consequences resulting from incorrect, incomplete, negligent or unsafe use of the installation system.

Please note: When handling and installing the LED modules, please operate carefully to avoid any bumps that may cause damage.